Name: ____________________________ Date: ________________ Period: _________

LO 3.9.A Inverse Trigonometric Functions

Key Vocabulary

- **arcsine:** ______________________________
  ____
- **arccosine:** ______________________________
  ____
- **arctangent:** ______________________________
  ____
- **restricted domain:** ______________________________
  ____
- **principal value:** ______________________________
  ____

Section 1 — Why Do We Need Restricted Domains?
The table below shows selected values of $f(x) = \sin x$.

x	$-\frac{\pi}{2}$	$-\frac{\pi}{6}$	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	π
$\sin x$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0

Notice that $\sin(\frac{\pi}{6}) = \sin(\frac{5\pi}{6}) = \frac{1}{2}$. This shows $\sin x$ is ________________ (circle: one-to-one / NOT one-to-one).

Because the sine function is periodic, it ________________________________ the horizontal line test, so it ________________________ have an inverse over its full domain.

Solution: Restrict the domain so the function becomes one-to-one.

Section 2 — Restricted Domains and the Inverse Trig Functions

Function	**Restricted Domain**	**Range**	**Inverse Name**	**Notation**
Sine	________________	$[-1, 1]$	Arcsine	$\arcsin x$ or $\sin^{-1} x$
Cosine	________________	$[-1, 1]$	Arccosine	$\arccos x$ or $\cos^{-1} x$
Tangent	________________	$(-\infty, \infty)$	Arctangent	$\arctan x$ or $\tan^{-1} x$

The **output** of an inverse trig function is an ________________ measure (the **principal value**).

Section 3 — Evaluating Inverse Trig Functions
Ask: "Which angle *in the restricted range* has the given trig value?"

Expression	**Think:**	**Answer**
$\arcsin\left(\frac{1}{2}\right)$	Which $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ has $\sin\theta = \frac{1}{2}$?	____________
$\arccos(0)$	Which $\theta \in [0, \pi]$ has $\cos\theta = 0$?	____________
$\arctan(1)$	Which $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ has $\tan\theta = 1$?	____________
$\arcsin\left(-\frac{\sqrt{2}}{2}\right)$	Which $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ has $\sin\theta = -\frac{\sqrt{2}}{2}$?	____________
$\arccos(-1)$	Which $\theta \in [0, \pi]$ has $\cos\theta = -1$?	____________

Guided Practice

Practice: Evaluate Evaluate each expression. Write the exact answer in radians.

1. $\arcsin(1) =$ ____________
2. $\arccos\left(\frac{\sqrt{3}}{2}\right) =$ ____________
3. $\arctan(-1) =$ ____________
4. $\sin^{-1}\left(-\frac{1}{2}\right) =$ ____________
5. $\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right) =$ ____________
6. $\tan^{-1}(0) =$ ____________